CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 8: **Life Processes in Animals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A person has a blocked bile duct. The food whose digestion will be most affected is: 1

(a) starch-rich food **(b)** protein-rich food
(c) water **(d)** fat-rich food

2. A man's heart rate is 80 beats per minute. In 24 hours, it beats: 1

(a) 80 times **(b)** 1,152 times **(c)** 1,15,200 times **(d)** 11,520 times

3. The reason that the human small intestine is long and coiled is: 1

(a) to store food
(b) to digest water
(c) to make a noise
(d) to provide a large surface area for absorption

4. A person has no pancreas. Which will be the most affected? 1

(a) digestion of water only **(b)** breathing
(c) absorption of oxygen **(d)** digestion of all three food types

5. A person breathes 16 times a minute, taking in 500 mL each time. The volume of air in one minute is: 1

(a) 8000 mL **(b)** 500 mL **(c)** 4000 mL **(d)** 16 mL

6. Herbivores have a longer small intestine than carnivores because: 1

(a) plant food takes longer to digest **(b)** they are slower
(c) they sleep **(d)** they are bigger

7. A person's stomach acid is removed. Which is the most likely effect? 1

(a) better fat digestion
(b) no change
(c) poor protein digestion and more germs reaching the intestine
(d) faster starch digestion

8. In one hour this volume is: 1

(a) 48 litres **(b)** 4800 litres **(c)** 480 litres **(d)** 8 litres

9. A carnivore's digestive tract compared to a herbivore's is: 1

(a) longer **(b)** absent **(c)** shorter **(d)** same

10. The villi are absent in the intestine. The consequence is: 1

(a) poor absorption of digested food **(b)** no excretion

(c) no breathing **(d)** no digestion

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain why a person feels warm after eating food rich in carbohydrates. 2

12. Explain why the right side of the heart has less thick walls than the left. 2

13. Why does oxygen move from the alveoli to blood? 2

14. Explain why the breathing and heart rates both rise during exercise. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the effects on the digestive system if the liver, pancreas and gall bladder are removed. 3

16. Explain why white blood cells are found in higher numbers at an infected wound. 3

17. Describe the role of fibre in preventing digestive disorders. 3

OR

A runner's pulse rate goes from 70 to 150 per minute. Explain what happens in the muscles and the blood. 3

18. Why does a diabetic person need to be careful about sugar-rich foods? 3

OR

A frog is placed in a closed jar of water without air for an hour. Explain what happens. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Design an investigation to find how the breathing rate changes with activity and present the data in a table with a conclusion. 5

OR

Explain how smoking and air pollution harm the respiratory, circulatory and excretory systems. 5

20. Describe the flow of blood from the heart through arteries, capillaries and veins and back, and say what is exchanged at the capillaries. 5

OR

Compare breathing in humans, fish and insects in terms of organs, medium and efficiency. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND